

Continuum Mechanics

class # 03

8/25/2026

Special Classes of Tensors

Definition: To any tensor $S \in V^2$ the associated tensor $S^T \in V^2$ defined as

$$\vec{v} \cdot (S\vec{u}) = \vec{u} \cdot (S^T\vec{v}) \quad \forall \vec{u}, \vec{v} \in V$$

is called the transpose (or adjoint) of S .

We say that S is symmetric (or self adjoint) iff
$$S^T = S.$$

We say that S is skew-symmetric iff
$$S^T = -S.$$

A tensor is called positive definite iff

$$\vec{v} \cdot (S\vec{v}) > 0 \quad \forall \vec{v} \in V.$$

A tensor is called symmetric positive definite iff it is (i) symmetric and (ii) positive definite.

Exercise: Show that the transpose is a uniquely defined object.

Definition: Let $S \in V^2$. We say that S is invertible iff there exists a tensor $A \in V^2$ such that

$$SA = AS = I.$$

In this case, we write $S^{-1} = A$.

Proposition: Let $S \in V^2$. If S is invertible, the inverse is unique. If S is invertible, S^T is also invertible and one finds that

$$(S^T)^{-1} = (S^{-1})^T.$$

Proof: Exercise.

Remark: We will use the unambiguous notation

$$S^{-T}$$

for

$$S^{-T} = (S^T)^{-1} = (S^{-1})^T.$$

Definition: A tensor $Q \in V^2$ that satisfies

$$Q^T = Q^{-1}$$

is called orthogonal.

If Q is orthogonal

$$QQ^T = Q^TQ = I.$$

Definition: An orthogonal tensor $Q \in V^2$ is called a rotation iff given any right-handed orthonormal frame

$$\mathcal{F} = \{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$$

The set

$$\mathcal{G} = \{Q\vec{e}_1, Q\vec{e}_2, Q\vec{e}_3\}$$

is also a right-handed orthonormal frame.

Proposition: Suppose that $S \in V^2$ and $\mathcal{F} = \{\vec{e}_i\}$ is a frame. Let $[S]$ be the matrix representation of S in frame $\mathcal{F}$. Then

$$[S^T] = [S]^T$$

If S is invertible, then

$$[S^{-1}] = [S]^{-1}, \quad [S^{-T}] = [S]^{-T}$$

Result (1.2): Let $S \in V^2$ be given. There exist unique tensors $E \in V^2$ and $W \in V^2$ such that

$$S = E + W$$

and $S^T = S$ and $W^T = -W$.

Proof: Set

$$E := \frac{1}{2}(S + S^T) \quad W := \frac{1}{2}(S - S^T).$$

Then $S = E + W$, clearly. Now, we need three facts that can be easily shown.

$$(A+B)^T = A^T + B^T, \quad \forall A, B \in V^2$$

$$(\alpha A)^T = \alpha A^T, \quad \forall A \in V^2, \alpha \in \mathbb{R}$$

$$(A^T)^T = A, \quad \forall A \in V^2.$$

If these are true, then

$$E^T = \frac{1}{2}(S^T + S) = E$$

and

$$W^T = \frac{1}{2}(S^T - S) = -W.$$

To prove uniqueness, suppose that there are two decompositions

$$E_1 + W_1 = E_2 + W_2$$

Then

$$\Delta := E_1 - E_2 = W_2 - W_1$$

We need a couple more facts suppose that $\alpha, \beta \in \mathbb{R}$ and $A, B \in V^2$ are symmetric (skew symmetric). Then $\alpha A + \beta B$ is symmetric (skew symmetric).

The only tensor that is simultaneously symmetric and skew symmetric is the zero tensor, $0 \in V^2$.

Thus $\Delta = 0 \in V^2$ and the result follows. //

Result (1.3): Given any skew-symmetric tensor $W \in V^2$ there is a unique vector $\vec{w} \in V$ such that

$$W\vec{v} = \vec{w} \times \vec{v}, \quad \forall \vec{v} \in V.$$

In any frame $\mathcal{F} = \{\vec{e}_i\}$

$$w_j = \frac{1}{2} \epsilon_{n,j,m} W_{n,m}.$$

Conversely, given any vector $\vec{w} \in V$, there is a unique skew-symmetric tensor $W \in V^2$ such that

$$\vec{w} \times \vec{v} = W\vec{v} \quad \forall \vec{v} \in V.$$

In any frame $\mathcal{F} = \{\vec{e}_i\}$, we have

$$W_{i,k} = \epsilon_{i,j,k} w_j.$$

Proof: ($\Rightarrow$) Existence. Existence is easy, for we are given the (possible) answer. Suppose the $\mathcal{F} = \{\vec{e}_i\}$ is our favorite frame. Let, as usual

$$[W]_{i,j} = W_{i,j}$$

Set

(1.9)

$$w_j = \frac{1}{2} \epsilon_{n,j,m} W_{n,m}, \quad j=1,2,3.$$

This defines

$$\vec{w} = w_1 \vec{e}_1 + w_2 \vec{e}_2 + w_3 \vec{e}_3.$$

We want to show that

(1.8)

$$W_{i,k} = \epsilon_{i,j,k} w_j$$

because this will imply, for arbitrary $\vec{v} \in V$

$$\begin{aligned} W_{ik} v_k &= \epsilon_{ij,k} W_j v_k \\ &= [\vec{W} \times \vec{v}]_i. \end{aligned}$$

We compute, using Result (1.1),

$$\begin{aligned} \epsilon_{ij,k} W_j &= \epsilon_{ij,k} \left(\frac{1}{2} \epsilon_{n,j,m} W_{n,m} \right) \\ &= \frac{1}{2} \epsilon_{ijk} \epsilon_{n,j,m} W_{n,m} \\ &= \frac{1}{2} (\delta_{in} \delta_{km} - \delta_{im} \delta_{kn}) W_{n,m} \\ &= W_{ik}. \end{aligned}$$

Thus (1.8) is true.

(Uniqueness): Suppose that there are two vectors that satisfy (1.8):

$$W_{ik} = \epsilon_{ijk} W_j$$

and

$$W_{ik} = \epsilon_{ijk} U_j$$

Then

$$\epsilon_{ijk} (W_j - U_j) = 0 \quad \forall \quad 1 \leq i, j \leq 3.$$

$$\epsilon_{ijk} = 1 \quad ijk = 123, 231, 312$$

$$ijk = 123: \quad W_2 - U_2 = 0$$

$$ijk = 231: \quad W_3 - U_3 = 0$$

$$i'jk = 312 \quad w_1 - u_1 = 0 \quad //$$

($\Leftarrow$): The other direction is in the book //

Definition: let $S \in V^2$

$$\text{sym}(S) = \frac{1}{2}(S + S^T)$$

$$\text{skew}(S) = \frac{1}{2}(S - S^T)$$

Suppose $W \in V^2$ is skew symmetric. Define

$$\vec{w} = \text{vec}(W)$$

as the unique vector given in Result (1.3). $\vec{w}$ is called the axial vector associated with $W \in V^2$.

Suppose $\vec{w} \in V$ is given. Define

$$W = \text{ten}(W)$$

as the unique tensor given in Result (1.3). W is called the axial tensor associated with $\vec{w}$.

Proposition: The functions $W = \text{ten}(\vec{w})$ and $w = \text{vec}(W)$ are inverses.

Proof: The proof is in the book and skipped for brevity.

Change of Basis

Definition: let $\mathcal{F} = \{\vec{e}_i\}$ and $\mathcal{F}' = \{\vec{e}'_i\}$ be two frames for $\mathbb{E}^3$. By the change of basis tensor from $\mathcal{F}$ to $\mathcal{F}'$ we mean $A \in V^2$ defined by

$$(1.11) \quad A = A_{ij} \vec{e}_i \otimes \vec{e}_j, \quad A_{ij} = \vec{e}_i \cdot \vec{e}'_j.$$

Proposition: let $\mathcal{F} = \{\vec{e}_i\}$ and $\mathcal{F}' = \{\vec{e}'_i\}$ be two frames. Then

$$(1.12) \quad \vec{e}'_j = A_{ij} \vec{e}_i, \quad 1 \leq j \leq 3$$

Similarly,

$$(1.13) \quad \vec{e}_i = A_{ik} \vec{e}'_k, \quad 1 \leq i \leq 3.$$

Proof: Observe that

$$\vec{e}'_j = c_1 \vec{e}_1 + c_2 \vec{e}_2 + c_3 \vec{e}_3$$

since $\mathcal{F} = \{\vec{e}_i\}$ is a basis. By orthogonality,

$$\vec{e}'_j \cdot \vec{e}_k = c_k.$$

Thus

$$\begin{aligned} \vec{e}'_j &= (\vec{e}'_j \cdot \vec{e}_1) \vec{e}_1 + (\vec{e}'_j \cdot \vec{e}_2) \vec{e}_2 + (\vec{e}'_j \cdot \vec{e}_3) \vec{e}_3 \\ &= (\vec{e}'_j \cdot \vec{e}_k) \vec{e}_k \\ &= (\vec{e}_k \cdot \vec{e}'_j) \vec{e}_k \\ &= A_{kj} \vec{e}_k. \end{aligned}$$

This (1.12).

Next,

$$\begin{aligned}\vec{e}_i &= (\vec{e}_i \cdot \vec{e}'_k) \vec{e}'_k \\ &= A_{ik} \vec{e}'_k,\end{aligned}$$

which is (1.13). //

Result (1.4): The components A_{ij} of the change of basis tensor A have the property that

$$\begin{aligned}A_{ij} A_{ik} &= \delta_{jk} \\ \text{and} \\ A_{ik} A_{jk} &= \delta_{il}\end{aligned}$$

In matrix notation,

$$[A]^T [A] = [I]$$

$$[A] [A]^T = [I]$$

Thus, the change of basis matrix is orthogonal.

Proof: Substitute (1.13) into (1.12) to get

$$\vec{e}'_j = A_{ij} A_{ik} \vec{e}'_k.$$

But

$$\vec{e}'_j = \delta_{jk} \vec{e}'_k$$

Thus

$$A_{ij} A_{ik} = \delta_{jk}, \quad 1 \leq j, k \leq 3.$$

Similarly, substituting (1.12) into (1.13) yields

$$e_i = A_{ik} A_{ek} \vec{e}_e$$

So

$$A_{ik} A_{ek} = \delta_{ie} \quad ///$$

Proposition: let $\mathcal{F} = \{\vec{e}_i\}$ be a frame and
 Suppose $\vec{v} \in V$ is arbitrary. Suppose in the
 frame

$$[\vec{v}]_{\mathcal{F}} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_{\mathcal{F}},$$

equivalently

$$\vec{v} = v_i \vec{e}_i.$$

let

$$[\vec{v}]_{\mathcal{F}'} = \begin{bmatrix} v'_1 \\ v'_2 \\ v'_3 \end{bmatrix}_{\mathcal{F}'}$$

be the representation in the frame $\mathcal{F}'$ and
 suppose that $A \in V^2$ is the change of basis
 tensor from $\mathcal{F}$ to $\mathcal{F}'$. Then

$$v_i = A_{ij} v'_j \quad 1 \leq i \leq 3$$

and

$$v'_k = A_{ik} v_i \quad 1 \leq k \leq 3$$

In matrix form

$$[\vec{v}]_{\mathcal{F}} = [A]_{\mathcal{F}} [\vec{v}]_{\mathcal{F}'}$$

and

$$[\vec{v}]_{\mathcal{F}'} = [A]_{\mathcal{F}}^T [\vec{v}]_{\mathcal{F}}$$

Where

$$[A]_{\mathcal{F}} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}_{\mathcal{F}}$$

Proof: A simple exercise. //

To simplify our notation, we will write

$$[\cdot] = [\cdot]_{\mathcal{F}}$$

and

$$[\cdot]' = [\cdot]_{\mathcal{F}'}$$

Traces, Determinants and Exponentials

Definition: Let $S \in V^2$ be arbitrary and suppose $\mathcal{F} = \{\vec{e}_i\}$ is a frame. Define $\text{tr} : V^2 \rightarrow \mathbb{R}$ via

$$\text{tr}(S) = \text{tr}([S]) = S_{ii}$$

where $[S]$ is the matrix representation in the frame $\mathcal{F}$. //

You might think that $\text{tr}(S)$ depends upon the frame we choose but this is not the case.

To see this we need a result.

Proposition: Suppose that $S \in V^2$ is arbitrary,
 $\mathcal{B} = \{\vec{e}_i\}$ and $\mathcal{B}' = \{\vec{e}'_i\}$ are two bases and

$$A = A_{ij} \vec{e}_i \otimes \vec{e}_j \quad A_{ij} = \vec{e}_i \cdot \vec{e}'_j$$

is the change of basis tensor from $\mathcal{B}$ to $\mathcal{B}'$. Then

$$[S] = [A][S']^T[A]^T$$

and

$$[S']^T = [A]^T[S][A]^T.$$

Proof: An exercise. ///

Result (1.5): Let $S \in V^2$ be given, with $[S]$ and $[S']^T$ the representations of S in bases $\mathcal{B}$ and $\mathcal{B}'$.
 Then

$$\text{tr}([S]) = \text{tr}([S']^T).$$

Proof:

$$\text{tr}([S]) = S_{ii}$$

$$= [A]_{ik} [S']_{kl}^T [A]_{li}$$

$$= [S']_{kl}^T [A]_{ik} [A]_{li}$$

$$= [S']_{kl}^T \delta_{kl} \quad \text{Result (1.4)}$$

$$= [S']_{kk}^T$$

$$= \text{tr}([S']^T) \quad ///$$

Definition: let $S \in V^2$ be given and suppose $\mathcal{H} = \{\vec{e}_i\}$ is a frame. The determinant $\det: V^2 \rightarrow \mathbb{R}$ is the mapping defined by

$$\det S = \det([S]) = \epsilon_{ijk} [S]_{i1} [S]_{j2} [S]_{k3}$$

Result (1.6): let $S \in V^2$ be arbitrary and suppose that $\mathcal{H} = \{\vec{e}_i\}$ and $\mathcal{H}' = \{\vec{e}'_i\}$ are two frames. Then

$$\det([S]) = \det([S]')$$

Proof: Exercise. ///

Proposition: let $S, T \in V^2$ be arbitrary. S is invertible iff

$$\det(S) \neq 0$$

and

$$\begin{aligned} \det(ST) &= \det(S) \det(T) \\ &= \det(TS) \end{aligned}$$

If $Q \in V^2$ is orthogonal then

$$|\det(Q)| = 1$$

If Q is a rotation then

$$\det Q = 1.$$

Definition: let $S \in \mathcal{V}^2$ be arbitrary.

$$\begin{aligned}\exp(S) &= I + S + \frac{1}{2}S^2 + \frac{1}{6}S^3 + \dots \\ &= \sum_{j=0}^{\infty} \frac{1}{j!} S^j. \quad //\end{aligned}$$

This series converges for any $S \in \mathcal{V}^2$

Result (1.7) let $W \in \mathcal{V}^2$ be any skew symmetric tensor. Then $\exp(W) \in \mathcal{V}^2$ is a rotation matrix. In particular,

$$(\exp(W))^T \exp(W) = I$$

and

$$\det(\exp(W)) = 1$$

Furthermore,

$$\exp(W)^T = \exp(W^T).$$

Proof: An exercise. //